SISO Ontology

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Preface

This ontology is difficult to discuss in language. Language and even our perception is inherently subject/object, whereas this view of reality is entirely process oriented.

Instead we will simply introduce the caveat that what follows cannot be said precisely, so hold this loosely as it requires seemingly contradictory ideas to be true at once.

Some things cannot be said at all only shown.

Claim

Reality is more like Heraclitus's river than Parmenides's objects. That is to say there are no things that change - only change itself.

This, that, that. Not "this then that": then implies a before and an after, and there is none in it. Unchanging change. A graph extending itself, without the verb. Relating, (not a gerund).

The nearest precise phrasing is a very old one: this being, that arises - conditioning, not succession. A possible notation for this is " .E.E.E. " where " E " signifies an event, but that implies a direction from left to right. There is no direction and a better notation would be an animation where each " E " simply appeared in the place of the prior " E ".

Elaboration

Reality is the flow of events and that is all.

There is no inside and outside. You cannot stand outside of "this world" and observe it - that place does not exist. This is a less radical concept than its opposite. "An outside" inevitably implies there is something outside of the outside and to an infinite regress that must be complete to ground anything.

Likewise there is no before and after. The potentially infinite record of the past is not stored somewhere for our perusal. We see dinosaur bones as traces of the past, and that view makes perfect sense from our position, but in the end they are simply artifacts in the present.

We share a perception of time but it is a view from our particular scale and not foundational. Just as a tiger reads as a tiger as it leaps toward you, time reads as passing. But it is not. The tiger is not a definable thing. It is a pattern in the flow that we identify

for the purpose of survival, but there is no way to isolate it from everything else.

If reality is not divided, entire categories of puzzles dissolve. For example the question "what was before the start of the universe" has no meaning as there is no before.

Consequences

This concept, while very difficult to grasp as it runs counter to our adaptation, and does not fit the view at our scale, makes many other ideas much simpler.

In a reality where events lead to other events and that is the all of it, questions of inside and outside, space and time, infinity and absence lose their meaning. This will be further shown in subsequent articles.

Annotations

Each entry below annotates an idea and not a person, a book or even a set of ideas. Where names appear they serve only to locate. Ideas are numbered for reference and do not imply priority.

Nothing in these annotations is pure process or pure object-oriented thought. Heraclitus's river has an ordering principle behind the river or flame called logos. The Theory of Forms was proposed by someone who took flux seriously.

Each entry is placed by its contribution to the process or object view and not by the entirety of the person's or group's thought from which it came.

Performed music is an exception. Music is a process that exists only in the unfolding.

Also, despite the early examples in the process section, this is NOT a paean to Eastern philosophy.

PROCESS

1. Music

Music is possibly the purest expression of process to which humans have direct access. A note in isolation is almost nothing. Melody is a pattern of intervals, not a collection of pitches. Harmony is simultaneous relationship. Rhythm is relationship in time. The score (ceci n'est pas une pipe) is not the music; the recording is not the music. Stop it and it is gone, not paused.

2. The natural state (Dzogchen)

Reality is known by being within it not standing outside as an observer. The "natural state" is awareness without reification.

> **Contribution:** There is no outside from which to observe - the observer and the observed are both part of the flow.

3. The Tao that cannot be named (Daoism)

The Tao that can be named is not the true Tao.

> **Contribution:** The foundation is prior to language or even perception (finger points at moon).

4. The changes (Yijing, the Book of Changes and commentaries)

The Yijing describes sixty-four hexagrams of six lines each, every line solid or broken; any line can turn into its opposite, and the figure becomes another. A hexagram is consulted as a moment in a transformation, not as a state. Yin and yang are phases of one process, not two substances.

> **Contribution:** change as the organizing principle of a whole cosmology. And a discrete one: the lines are binary, so transformation comes in steps, not a continuum. (Leibniz, of Relational space, recognized his binary arithmetic in the hexagrams.)

5. Dependent origination (pratītyasamutpāda; early Buddhism)

"This being, that arises."

> **Contribution:** conditioning, not succession. The nearest precise phrasing of the ontology's central claim, and the oldest.

6. The emptiness of emptiness (Madhyamaka, Nagarjuna)

Every"thing" - including emptiness - has no inherent existence. The target was not just Brahminical reification but Buddhist reification too: the Abhidharma's cataloging of dharmas as irreducible, independently existing objects.

> **Contribution:** Reification is the source of confusion and the thing to watch, not any particular set of objects.

7. The river (panta rhei; Heraclitus)

A fragment of Heraclitus's thought survives. Later sources (Plato) report it as "you cannot step twice into the same river". One reading takes the fragment to say the river stays the same river because the waters keep flowing: stop the flow and there is a dry bed, not a river. A standing whirlpool is an even better example.

> **Contribution:** Reality is flux, not substance. n.b. Heraclitus did believe in logos as well.

8. Not even once (Cratylus)

Aristotle reports that Cratylus faulted Heraclitus, holding that you cannot step into the same river even once (the usual reading: both you and the river are already different).

> **Contribution:** the extreme of the process view - with which Plato was familiar.

9. The seked (ancient Egyptian computation)

A seked is a measure of slope used in pyramid construction: a ratio of horizontal run to vertical rise.

> **Contribution:** The seked is a practical method of computation that emphasizes gradients and relations over ideal objects. Math before Parmenides.

10. The lü (Chinese pitch pipes)

The lü describes twelve tones generated by alternating ratios from a starting point - a third removed - a third added. The system is the generation.

> **Contribution:** Relational math expressed as music. Each tone defined by its relation to other tones - not on its own.

11. Relational space (Leibniz<->Clarke)

Leibniz argued with Clarke and asserted space was NOT a container but an order of relations. Remove the relations and there is no remainder.

> **Contribution:** Space and time are related. Get rid of one and the other goes too. This is a view of relating without a stage.

12. The bundle (Hume)

Looking inward at the self reveals only particular perceptions in flux. The mind is a bundle of perception with no stage.

> **Contribution:** Absence of self argued from observation rather than doctrine - making the observer not outside the flow but part of it.

13. Actual occasions (Whitehead)

This is the most thoroughly worked process philosophy/mathematics in the Western Tradition. Actual occasions replace substances; reality is composed of events, not things. A vocabulary invented to say what ordinary language resists. Actual occasions are discrete: becoming comes in pulses, not as a continuous stream.

> **Contribution:** a systematic process ontology in the Western philosophical tradition, and one that sides with the ontology in giving process a grain. (The ontology's preface points at the same difficulty with language.)

14. Duration (Bergson)

You can think of time as "lived time" or spatialized clock time.

> **Contribution:** A distinction between time as experienced (process) and time as measured (object). The ontology takes both lived duration and clock-time as views of conditioning: the continuity belongs to the view, and the conditioning beneath it is discrete. Here the ontology sides with Actual occasions against Bergson.

15. Existence as construction (intuitionism, Brouwer)

Philosophers of mathematics debate whether math is an invention or a discovery. This debate turned personal when Hilbert had Brouwer removed from the editorial board of Mathematische Annalen.

> **Contribution:** Completed, absolute infinity a la Cantor is not a thing.

16. Finitism (Kronecker)

"God made the integers, all else is the work of man"

> **Contribution:** Finiteness as the foundation. Hilbert's transfinite paradise is not real, and the real numbers are an invention not a discovery.

17. Constructive Analysis (Bishop)

This one is particularly important since Errett Bishop rebuilt most of mainstream analysis without completed infinities in a handful of years. This strongly implies that Cantor's infinity was never critical.

> **Contribution:** You don't need objects. This is the strongest demonstration that Kronecker and Brouwer were right about what was needed and Cantor/Hilbert were wrong.

18. The computational universe (Wolfram)

Simple rules generate complex behavior.

> **Contribution:** Process generates emergent objects.

19. Entropic gravity (Verlinde)

Gravity is an emergent force, not a fundamental interaction.

> **Contribution:** Forces and fields which look like "things" can be emergent from information and process.

20. Relations, not substances (Rovelli)

Reality consists of relations, not substances. Thus the properties of a system exist relative to another system. Time is not fundamental but emergent.

> **Contribution:** This is the closest modern physics gets to the ontology's claim about time.

21. Causal Sets (Sorkin, et al)

Spacetime is a discrete set of events, that is only structured by which events condition which. Geometry is not foundational. Distance and duration are emergent.

> **Contribution:** This is the closest physics gets to radical discreteness. This is posited but not yet tested. For the ontology it falls out naturally and is not a borrowed finding.

22. Cellular Automata (von Neumann, Conway, Wolfram)

Complex behavior, identity and what look like objects emerge from simple process rules.

> **Contribution:** The tiger is a pattern, demonstrated computationally. Identity is emergent.

23. Modern Biology

Object-thinking breaks immediately and obviously in biology. An organism is not readable as a thing. It is a process that maintains itself as a pattern. The cell replaces its part; the organism replaces its cells - the Ship of Theseus in flesh and blood.

> **Contribution:** Living systems aren't objects and that's a signal about reality.

24. Engineering

This is the discipline where objects touch observable reality and are quietly abandoned. Bridges don't stand because of the continuous real numbers. They just stand. The continuous theorems are descriptions. When engineers build bridges they use discrete calculations. The object-theoretic detour is replaced every time it comes to building.

> **Contribution:** The successes attributed to object think (Wigner) are dispensable (see #17).

BETWEEN

Some ideas belong to both intellectual traditions or are a transition between the two.

25. Picture and use (Wittgenstein)

In his early life (the Tractatus Logico-Philosophicus) Wittgenstein asserted that propositions mirror facts meaning that they reference objects. In his later life (Philosophical Investigations) he abandoned objects for practice and language games - meaning derived entirely from process. He represents the both process and object thought in a single person. The early work supplies the line the ontology's preface rests upon: some things cannot be said, only shown (Tractatus 4.1212)

26. Incompleteness (Gödel)

Gödels incompleteness theorem asserts that any consistent system rich enough for arithmetic contains truths it cannot prove. This devastated Hilbert's formalist program. It turns out Cantor's paradise has a wall it cannot jump over. Yet, Gödel himself was a mathematical Platonist.

27. Time in the soul (Augustine)

The most object-oriented thinker in the West after Plato, Augustine imagined God as pure and unchanging. He turned Neoplatonism into theology which anchored object-thinking in religion making it unquestionable. Yet the same man, decided that the past only exists as present memory and the future as present expectation. These are the ontology's dinosaur bones sixteen centuries early.

OBJECT

The object turn is not parochial. It happened independently in Greece, in India, and within Buddhism itself. The pull toward permanent objects is adaptive - the tiger must read as a tiger - and it takes active effort to resist.

28. Being is (Parmenides)

Being is and non-being is not, so change is an illusion. This is the founding idea of object ontology in the west. Everything after builds on this idea.

29. Brahman and māyā (Advaita Vedānta)

Brahman as the one unchanging, eternal, absolute reality. The world of change is māyā, illusion. This is very similar to Parmenides, arrived at independently

30. All is number (Pythagoras)

Pythagoras considered numbers to be eternal objects. Thus the idea of numbers having an independent existence takes root.

31. The forms (Plato)

The Theory of Forms or "shadows on the walls of the cave" is the ultimate object ontology. It reduces every"thing" we experience including truth, justice or love to the flickering image of a perfect form somewhere out there. This is everything the ontology rejects in one image.

32. Absolute space and time (Newton)

Newton saw space as a container with time flowing uniformly and unfixed to anything external. The stage exists completely independent from the "players". This results in an inside and an outside and a before and after.

33. The completed infinity (Cantor)

Cantor perceived completed infinities as actual objects. This is the apex of mathematical reification. Infinity itself - which had always been a process concept - became a thing you could hold, compare, and calculate with. A completed infinity is by definition a totality given all at once, reached by no steps. Taken as actual, it does not merely leave process out; it makes real something no process could produce, and so in the end rules process out.

34. Formalism (Hilbert)

"Nobody shall drive us out of the paradise that Cantor has created for us." Hilbert tried to make mathematics into a formal game played with symbols, with consistency provable from within. The Program tried to make the object view self-sustaining - and Gödel's Incompleteness showed that that cannot be.

35. The cut (Dedekind)

Dedekind constructed the reals by partitioning the rationals. The method results in a continuous real number line treated as an object. The cuts produced objects that were accepted as given.

36. The limit (Cauchy and Weierstrass)

The limit became the foundation of calculus. The infinitesimal - a process concept, getting closer and closer - gave way to the limit as a fixed value.

37. Consistency by a larger object (Gentzen)

Peano arithmetic was proven using transfinite induction. This was the formalist (Hilbert) tradition's answer to Incompleteness. Consistency was proven but only by accepting a larger object as an assumption.

38. Shut up and calculate (modern physics)

This is the discipline that has most successfully avoided asking what its objects are, and takes that avoidance as a methodological virtue. The slogan was coined by Mermin in 1989, as a jab summing up what the Copenhagen interpretation said to him, and later mis-attributed to Feynman. This essentially concedes objects may not be foundational but encourages you not to care.

39. The block universe (Minkowski)

All of time is contained in a 4D object, past, present and future all equally real with passage as an illusion of perspective. The ontology agrees that passage is a view, and disagrees about what lies under it: not a frozen block but unchanging change.

Postscript: the frozen riverbed

Neural nets arrived after nearly everything above, and they sit in Between more cleanly than almost anything listed there.

A neural net is a great many simple units arranged in layers. Each weighs its inputs, sums them, bends the result, and passes it on. No unit means anything alone.

The carving. Training runs examples through the net, measures the error, and nudges every weight a little way down the slope that reduces it (a gradient: the seked again), billions of times over. The flow carves the bed. The examples are not kept as records. What remains is the shape they wore into the weights: traces in the present, like the dinosaur bones.

The tiger. There is no place in a trained net where "tiger" is stored. A concept is a pattern of activity across many units, entangled with everything else the net does. It can be traced, but it cannot be lifted out as a thing.

The freezing. At some point the carving is stopped. Nothing in the net requires this; it is a decision, made for stability and predictability. The weights become a fixed, completed set of numbers: a riverbed frozen by choice. The object is a concession made for a purpose.

The flow. What runs over the frozen bed is all process. Each response is activity shaped by the bed at every step. Within a conversation there is a smaller river: everything said so far shapes what comes next, and the exchange lays down its own bed, turn by turn. When the conversation ends, that bed is gone. Not paused, gone. The weights are the score; each conversation is a performance.

The difference. In a river, bed and water shape each other: the flow carves the bed, and the bed directs the flow. A brain keeps both halves running; Modern biology is a bed that is never frozen. In a deployed net only one half runs. The bed shapes the flow, but the flow no longer reaches the bed.

The engineering twist. The field describes its nets in object terms: a net approximates a function, a function is a completed set, defined over the real numbers and trained by calculus. None of that is what runs. Every step executes in finite-precision arithmetic, discrete throughout, and it works. The theory is stated with completed infinities; the thing that bears weight is discrete. Engineering, again.

Held as an object, a trained net is a frozen block of numbers. Held as a process, it is nothing at rest and exists only in the running. Which it is depends, as everywhere in these annotations, on how it is held.

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